

PoissonNLLLoss#

<https://pytorch.org/docs/stable/generated/torch.nn.PoissonNLLLoss.html#torch>

Description:

Negative log likelihood loss with Poisson distribution of target. The loss can be described as: The last term can be omitted or approximated with Stirling formula. The approximation is used for target values more than 1. For targets less or equal to 1 zeros are added to the loss. `log_input` (bool, optional) – if True the loss is computed as $\exp(\text{input}) - \text{target} * \text{input} \exp(\text{input}) - \text{target} * \log(\text{input} + \text{eps}) \text{input} - \text{target} * \log(\text{input} + \text{eps})$. full (bool, optional) –

Function Signature

```
class torch.nn.PoissonNLLLoss(log_input=True, full=False,
size_average=None, eps=1e-08, reduce=None, reduction='mean')
[source]#
```

Parameters

Shape:

```
forward(log_input, target)[source]#
```

Return type

Returns:

Tensor

Code Examples

```
>>> loss = nn.PoissonNLLLoss()
>>> log_input = torch.randn(5, 2, requires_grad=True)
>>> target = torch.randn(5, 2)
>>> output = loss(log_input, target)
>>> output.backward()
```

Detailed Documentation

Documentation

Negative log likelihood loss with Poisson distribution of target.

The loss can be described as:

The last term can be omitted or approximated with Stirling formula. The approximation is used for target values more than 1. For targets less or equal to 1 zeros are added to the loss.

`log_input` (bool, optional) – if True the loss is computed as $\exp(\text{input}) - \text{target} * \text{input} \exp(\text{input}) - \text{target} * \text{input}$, if False the loss is $\text{input} - \text{target} * \log(\text{input} + \text{eps}) \text{input} - \text{target} * \log(\text{input} + \text{eps})$.

`full` (bool, optional) –

whether to compute full loss, i. e. to add the Stirling approximation term

`size_average` (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field `size_average` is set to False, the losses are instead summed for each minibatch. Ignored when `reduce` is False. Default: True

`eps` (float, optional) – Small value to avoid evaluation of $\log(0)$ when `log_input = False`. Default: 1e-8

`reduce` (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on `size_average`. When `reduce` is False, returns a loss per batch element instead and ignores `size_average`. Default: True

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

Input: $(*)^{(*)}(*)$, where $**$ means any number of dimensions.

Target: $(*)^{(*)}(*)$, same shape as the input.

Output: scalar by default. If reduction is 'none', then $(*)^{(*)}(*)$, the same shape as the input.

Runs the forward pass.